SUWARNA PYAKUREL

Engineering Undergraduate | Applied AI, Software & Technical Leadership

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Profile

Electronics, Communication and Information Engineering undergraduate with projects in explainable deep learning, signal processing, IoT and software systems. Co-founder of Curio Tech, combining hands-on engineering with operations, institutional partnerships and technical outreach. Experience delivering event technology for international STEM programs and supporting student learning communities.

Education

Tribhuvan University, Institute of Engineering Sep 2023 – Expected 2027
B.E. in Electronics, Communication and Information Engineering | Purwanchal Campus, Dharan

Coursework: programming, algorithms, electronics, communication systems, embedded systems, signal processing and artificial intelligence.

SOS Hermann Gmeiner Secondary School Mar 2020 – Jul 2022
Higher Secondary Education, Science | GPA: **3.75** | Bhaktapur, Nepal
Head Boy, batch 2078 BS; participated in STEM and voluntary activities.

Technical & Research Projects

Explainable Pneumonia Detection & Severity Estimation 2026
Academic minor project, IOE Purwanchal Campus | 4-member team | Source code

- Co-developed a chest X-ray research pipeline for Normal/Pneumonia classification, integrating training, evaluation, explainability and web inference; selected **EfficientNet-B0** with **ROC-AUC 0.9624** and **F1-score 0.9185** in project evaluation.
- Integrated **Grad-CAM** heatmaps and a post-prediction severity heuristic combining model confidence with activation extent; connected image upload, inference, visualization and report export.

Tools: Python, PyTorch, CNNs, EfficientNet-B0, Grad-CAM, FastAPI, Express, React

Motor Vibration Fault Detector Mar 2025
Signal processing and predictive maintenance | Source code

- Built FFT-based diagnostic logic for rotor imbalance, shaft misalignment and bearing faults using RMS, kurtosis, crest factor, peak-to-peak amplitude, spectral energy and harmonic features.
- Implemented a **0–100 equipment health score**, diagnostic reports and maintenance recommendations; validated core logic through **22 automated tests**.

Tools: Python, NumPy, FFT, feature engineering, time/frequency-domain analysis

QR-Based Delegate Tracking & Activity Monitoring Oct 2024
Independent event technology project | Kathmandu, Nepal | IOAA-Jr. 2024

- Independently designed and deployed QR attendance and activity tracking for **200+ participants, delegates, organizers and personnel**, including international delegates from **19 countries**.
- Centralized checkpoint check-ins, attendance records and activity logs; managed deployment, on-site technical support, monitoring and troubleshooting.

Tools: Python, Django, SQLite, JavaScript, QR encoding, data logging

IoT-Based Smart Agriculture System Apr – May 2023
4-member team | Robotics Association Nepal, Kathmandu | Yantra 9.0 winner, 2024

- Co-developed an ESP8266 NodeMCU prototype using soil-moisture, DHT11, BMP280 and rain sensors; automated irrigation-pump and cooling-fan control through sensor thresholds.
- Added OLED monitoring over I2C and ThingSpeak logging; led Team ERC's winning Business Cup Expo presentation at Yantra 9.0 in 2024.

Tools: ESP8266, Arduino, sensors, IoT, I2C, ThingSpeak, actuator control

Experience & Leadership

Curio Tech Pvt. Ltd. Oct 2025 – Present
Co-Founder & Chief Operating Officer | Nepal

- Lead operational execution, strategic partnerships, institutional collaborations and financial coordination for a technology enablement startup focused on digital transformation.
- Coordinate project delivery across technical teams and external stakeholders; oversee PR, outreach and the company's digital presence.

EXCESS — IOE Purwanchal Campus Feb 2026 – Present
Advisor & DataCamp Maintainer | Dharan, Nepal

- Advise student-led technical programs, support transitions between organizing teams and maintain DataCamp learning access for programming and data-skills development.
- Previously served as **PR & Outreach Lead (Jul 2025 – Feb 2026)**, coordinating society communications, visibility and student engagement.

ICRTAI — International Conference on Recent Trends in AI Apr 2025 – Present
Outreach Coordinator | Birtamode, Nepal

- Coordinate media relations, targeted campaigns and stakeholder communications connecting researchers, academics, students and technical communities around the 2025 conference initiative.
- Work with organizing committees to align outreach, participation and research-community engagement with conference objectives.

CS50 AI Nepal — EXCESS Oct 2024 – Present
PR & Outreach Lead | IOE Purwanchal Campus

- Lead program communications, outreach campaigns and partnerships; support AI learning sessions, workshops, event planning and student engagement.

Nepal Astronomical Society Oct 2023 – Present
Undergraduate Student Member | Nepal

- Served as **Country Guide & Tech Lead, IOAA-Jr. 2024**, supporting international participant coordination, event logistics and technical operations.
- Organized NASA Space Apps Challenge activities; contributed to astronomy outreach and mentoring, working with Biratnagar and Koshi Province stakeholders on scientific education.

Astronomy and Space Science Study Research Committee Jun 2023 – Present
Member | Biratnagar Metropolitan City

- Support astronomy education and youth engagement through Python, Linux and scientific-computing workshops; contribute to policy drafting and program planning with local government.

Municipal Child Club Network Jul 2019 – Mar 2021
Secretary; subsequently Advisor (2023–2024) | Biratnagar, Nepal

- Coordinated children's engagement and development initiatives, community programs and stakeholder communication; subsequently advised the network's activities.

Additional technical-community leadership: Executive leadership with Robotics Club, IOE Purwanchal Campus; involvement in Koshi Mini Yantra 2.0 and regional STEM initiatives.

Technical & Professional Skills

Programming & software: Python, C, C++, JavaScript, OOP, DSA, Django, FastAPI, React, Express, SQLite, Pygame, web scraping, frontend development, Git/GitHub, Linux; exploring Go.

AI & signal processing: PyTorch, deep learning, CNNs, explainable AI, Grad-CAM, model evaluation, NumPy, FFT, feature engineering, optimization and data visualization.

Embedded systems & IoT: ESP8266/NodeMCU, Arduino, sensor integration, I2C, ThingSpeak, automation and robotics.

Research: Literature review, problem formulation, experimentation, critical analysis, research prototyping and technical documentation.

Consulting & leadership: SAP ERP, process analysis, requirements gathering, user support, operations, partnerships, strategic branding, PR, event management and public speaking.

Software Products & Additional Projects

iLoveQR.app — QR Generation & Design Platform Deployed product

Co-developer | 4-member team | Web application

- Co-developed a QR design platform supporting **10 content types**: URLs, vCards, Wi-Fi, text, email, SMS, phone calls, crypto payments, social profiles and calendar events.
- Added **9 visual presets**, configurable colors, logos, frames and shapes, live previews, and PNG/SVG/JPEG/PDF export with selectable resolutions up to **2048 px**.

Markdown PDF Studio 2026

Built a lightweight desktop Markdown editor with live preview, syntax highlighting, footnotes, HTML sanitization and A4 PDF export using Python, PySide6, markdown-it-py and Qt WebEngine.

Travelling Salesman Problem Algorithm Visualizer 2025

Implemented **Brute Force, Genetic Algorithm and Simulated Annealing** in Python/Pygame to explore an NP-hard optimization problem; enabled interactive city placement, live route evolution and comparison of computational trade-offs. Academic project, IOE Purwanchal Campus, Dharan.

AVL Tree Visualizer 2025

Developed a Python visualization of self-balancing binary search trees, demonstrating insertions, rotations, height updates and balancing behavior as part of DSA lab work.

Personalized Budget Tracker Mar 2025

Built a Python desktop application for income and expense tracking, transaction management, spending-pattern visualization and data export through a graphical interface.

Love Note Generator Mar 2025

Created a playful Python application with randomized romantic notes, interactive challenges and a gamified "Love Meter."

Study Tracker Feb 2025

Developed a Django application with user authentication, subject and exam-date management, unit-wise progress tracking, visual progress charts and syllabus-file uploads.

Library Management System Mar 2024

Built an object-oriented C++ application for library-resource management and book tracking. Academic project at IOE Purwanchal Campus, supervised by Er. Tantra Nath Jha.

Bank Management System Jan 2024

Developed a C application for account management, transactions and data handling. Academic project at IOE Purwanchal Campus, supervised by Er. Pravin Sangraula.

Awards & Recognition

- **Yantra 9.0 — International Robotics Competition (2024):** Part of Team ERC, winner of the **Swarmanoid** and **Business Cup Expo** titles; led the Business Cup Expo presentation.
- **X-Hack — Best Idea Presentation Award:** Recognized for an innovative project concept at a hardware hackathon in Nepal.
- **Main-belt asteroid 2020 WX9:** Received a provisional-discovery certificate through an asteroid-search campaign.

Research Interests

Explainable AI, medical AI, scientific machine learning, signal processing, predictive maintenance, IoT and embedded intelligence, data-driven systems, and robotics.

Languages

Nepali — Native | English — Highly proficient | Hindi — Highly proficient